

Calculator Model / No.

Name _____ Class: _____ Reg Number: _____

MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2**CHEMISTRY****9647/02**

Paper 2 Structured Questions

19 September 2011**2 hours**

Additional Materials: Data Booklet

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.
Write your calculator brand and model/number in the box provided above.

Answer **all** questions in the spaces provided on the question paper.

All working must be shown clearly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE	
Q1	/ 12
Q2	/ 17
Q3	/ 21
Q4	/ 10
Q5	/ 12
Total	/ 72

Answer **all** questions in the space provided.

1 Planning (P)

You are provided with solutions **FB 3**, **FB 4** and **FB 5**.

FB 3 and **FB 4** are either
1.0 mol dm⁻³ nitric acid **or**
2.0 mol dm⁻³ ethanoic acid

whereas **FB 5** is 1.0 mol dm⁻³ sodium hydroxide.

You are to plan experiments that will enable you to identify the solution that matches each of **FB 3** and **FB 4**, hence, calculate $\Delta H_{\text{neutralisation}}$ for the reaction between ethanoic acid and sodium hydroxide.

You are provided with the following apparatus:

-10 °C to 110 °C thermometer,
100 cm³ beaker,
50 cm³ measuring cylinder

The measuring cylinder should be rinsed thoroughly before each use.

- (a) (i) When determining the identity of the two acids listed above,

the independent variable is

the dependent variable is

- (ii) Plan an experiment to find which of the solutions is 1.0 mol dm⁻³ nitric acid and which is 2.0 mol dm⁻³ ethanoic acid.

Your plan must identify the acids solely based on the change in temperature. Mathematical processing of the temperature change is thus **not** required.

Your experiment(s) must use only the solutions and apparatus above.

Your plan should give a step by step description of the method including:

- the apparatus used for measurement
- appropriate volumes of reagents
- how you would measure the various variables

[4]

- (b)** Explain clearly how you could determine the identity of the acids using the difference in temperature rise in the experiment proposed in **(a)**.

[2]

- (c)** Show the mathematical expression for the enthalpy change of neutralization of sodium hydroxide, using the volume proposed in **(a)** with 25 cm³ of ethanoic acid.

The temperature change measured in **(a)** should be represented by ΔT .

[4.2 J of heat energy raised the temperature of 1 cm³ of any solution by 1°C.]

[2]

(d) A student accidentally diluted both the ethanoic acid and nitric acid prior to conducting the experiment above.

- (i) With a aid of a pH meter, a titrimetric method could still be used to differentiate the diluted ethanoic acid from the diluted nitric acid, even though their exact concentrations are not known.

Briefly describe the measurements to be taken, and how the measurements may lead to the identification of the two acids.

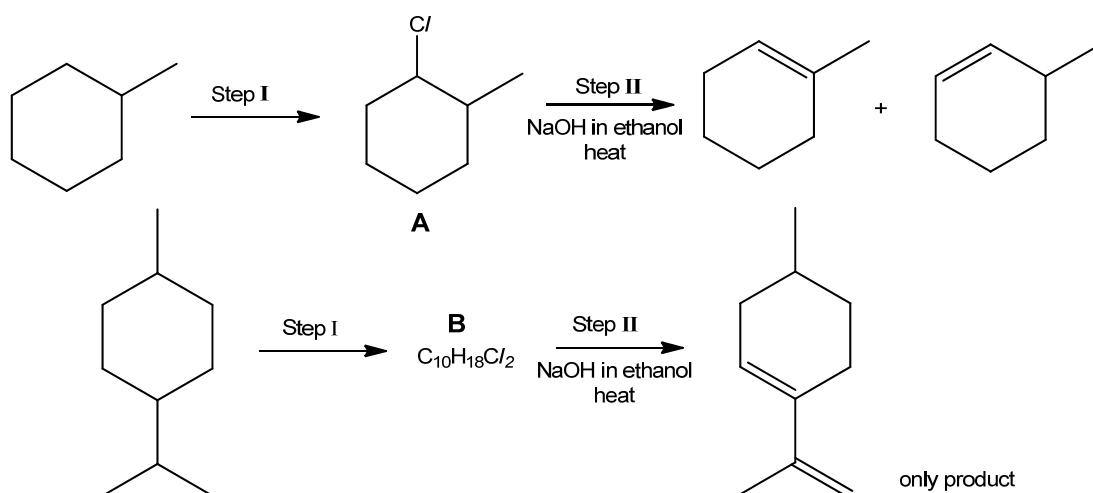
Precise details of how a titration is performed, along with the apparatus used are not required, though the identity of any indicator (if used) should be clearly stated.

- (ii) Other than the fact that a titrimetric method is more effective in the determination of the identity of a weak or strong acid when the concentration is unknown, compare the general reliability of using a titrimetric and thermochemical method in the determination of the identity of a weak or strong acid.

[4]

[Total: 12]

- 2(a)** Alkenes are very useful compounds, which are used as fuel and in the manufacture of a wide variety of polymers.



- (i)** Name and outline the mechanism in Step I to form **A**.

(ii) Draw the structure of **B**.

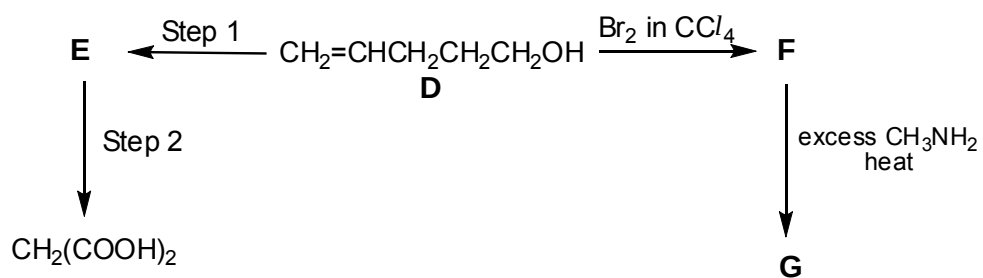
(iii) When $\text{C}_{10}\text{H}_{18}\text{Br}_2$ reacts with hot NaOH in ethanol in Step II, the rate of reaction is found to be faster. Explain why.

(iv) When **B** reacts with hot NaOH in ethanol in Step II, another side-product, **C** ($M_r = 228.0$), is formed.

Suggest the structure of **C**.

[7]

(b) The flow scheme below involves the reaction of **D**.



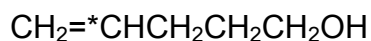
(i) Draw the structures for **E**, **F** and **G** in the boxes provided below.

E	F
G	

(ii) State the reagents and conditions for **Step 1** and **2** in the spaces provided.

	Reagents	Conditions
Step 1		
Step 2		

- (iii) **G** contains a chiral carbon but the eventual reaction mixture was found to be optically inactive. Suggest a reason for this observation.
- (iv) State the type of hybridisation with respect to the carbon atom indicated below by the asterisk * in compound **D**.



Sketch the shape of the hybrid orbitals of this carbon and indicate its bond angle.

- (v) It was found that when **D** is reacted with bromine, **H** ($\text{C}_5\text{H}_9\text{BrO}$) was formed.

Suggest the structural formula of **H**.

[10]

[Total: 17]

- 3** Old mine workings are often the cause of the pollution of mountain streams. One of the main chemical culprits of such pollution is iron pyrites, FeS_2 , in which iron has an oxidation state of +2.

The early miners unwittingly accelerated the chemical processes by grinding up unwanted rock containing some ore and dumping it in waste heaps.

- (a)** Oxygen in the air and rain water combined to oxidise the sulfide ions in solid iron pyrites to form an acidic solution of iron(II) sulfates.

Write a balanced ionic equation for this reaction.

[1]

- (b)** The resulting ions then run off into the local streams where the iron(II) ions are oxidised.

- (i)** By considering the appropriate E° values in the *Data Booklet*, explain why the water in the streams must be acidic for oxygen to oxidise the iron(II) ions.

- (ii)** Hence, write an equation for the oxidation of iron(II) ions in the stream water under acidic conditions.

[4]

(c) Aqueous iron(III) ions are present in the stream as *complex ions*.

(i) What is meant by the term '*complex ion*'?

(ii) Draw a diagram to show the shape of the aqueous iron(III) complex ions.

[2]

(d) The iron(III) ions are hydrolysed in the stream to form a precipitate which is basic in nature. The precipitate is referred to as '*yellow boy*' by the miners.

(i) Suggest the formula of the precipitate which gives rise to the yellow colour of '*yellow boy*'.

(ii) Write an equation to show the hydrolysis of aqueous iron(III) ions in water to form the '*yellow boy*', given that the final solution is acidic in nature.

[2]

- (e) As a result of the increased acidity of the stream water, other d-block metal ions are leached into the water. Zinc(II) ions are often found in the water but are not noticed because they are colourless.

(i) Suggest a reason why zinc(II) ions are colourless.

(ii) Explain why iron compounds are usually coloured.

[4]

- (f) Very dilute solutions of iron(III) ions are almost colourless. Addition of potassium thiocyanate, KSCN, leads to a deep red coloured complex.

(i) What type of reaction has occurred?

(ii) Suggest a technique (other than titration) you could use to determine the concentration of the complex formed.

[2]

- (g) When potassium cyanide is added to stream water, iron(III) ions form an orange red complex, $[\text{Fe}(\text{CN})_6]^{3-}$.

In an experiment, the standard electrode potential of the $[\text{Fe}(\text{CN})_6]^{3-} | [\text{Fe}(\text{CN})_6]^{4-}$ system is measured.

- (i) By means of a fully labelled diagram, show how the standard electrode potential of the $[\text{Fe}(\text{CN})_6]^{3-} | [\text{Fe}(\text{CN})_6]^{4-}$ system can be measured.

- (ii) The presence of CN^- significantly decreases the oxidising ability of an acidic solution of Fe(III) ions. Illustrate this effect by selecting appropriate $E^\ominus$ values from the *Data Booklet*.

- (iii) From the list of standard electrode potentials in the booklet, identify a gas that would reduce $[\text{Fe}(\text{CN})_6]^{3-}$ to $[\text{Fe}(\text{CN})_6]^{4-}$.

Explain your answer.

[6]

[Total : 21]

- 4 Silver ions are used to make silver ethanoate ($\text{AgC}_2\text{H}_3\text{O}_2$). It is an important ingredient in the manufacture of chewing gum and lozenges to aid cessation of smoking as it produces a repulsive taste when combined with cigarette smoke.

The solubility product for silver ethanoate can be found by the following method.

Preparation of saturated silver ethanoate solution by precipitation from solutions of salts

1. Using a burette, transfer 40.00 cm^3 of 0.20 mol dm^{-3} silver nitrate into a 100 cm^3 beaker.
2. Using another burette, transfer 40.00 cm^3 of 0.20 mol dm^{-3} sodium ethanoate into the beaker containing the silver nitrate.
3. Keep the contents well shaken.
4. Leave the mixture to stand for some hours at a temperature of 25°C .
5. Filter the mixture into a clean, dry 250 cm^3 beaker.

Determination of $[\text{Ag}^+]$ by titration with standard KSCN solution

6. Pipette 20.0 cm^3 of the saturated solution into a 250 cm^3 conical flask.
7. Add 6 drops of nitric acid into the conical flask.
8. Add 6 drops of iron (III) solution to act as an indicator.
9. Keep the contents well shaken
10. The mixture was then titrated against 0.10 mol dm^{-3} potassium thiocyanate.
11. The end-point being when the titration mixture turned red.

The volume of aqueous potassium thiocyanate required was 14.90 cm^3 .

- (a) Explain why the mixture was initially kept at 25°C (Step 4). [1]
- (b) Explain why there is a need to filter the mixture (Step 5). [1]
- (c) State the purpose of adding dilute nitric acid to the solution (Step 7). [1]
- (d) The end point of the titration was obscured by the formation of a white precipitate (Step 11).
State the identity of the white precipitate and write an equation with state symbols for its formation. [2]
- (e) (i) Calculate the number of moles of KSCN required to react with Ag^+ .
- (ii) Calculate the concentration of Ag^+ in the filtrate.
- (iii) Hence, calculate the solubility product of $\text{AgC}_2\text{H}_3\text{O}_2$.

- (iv) When a precipitate is formed, $\Delta G^\ominus_{\text{ppt}}$ in J mol^{-1} is given by the following expression:

$$\Delta G^\ominus_{\text{ppt}} = 2.303 RT \lg K_{\text{sp}}$$

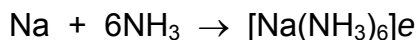
Given that the standard enthalpy change of formation of the precipitate of $\text{AgC}_2\text{H}_3\text{O}_2$ is $-17.5 \text{ kJ mol}^{-1}$, and using your answer in (iii), calculate $\Delta S^\ominus_{\text{ppt}}$ for the formation of the $\text{AgC}_2\text{H}_3\text{O}_2$.

- (v) Explain the significance of the sign of your answer in (iv).

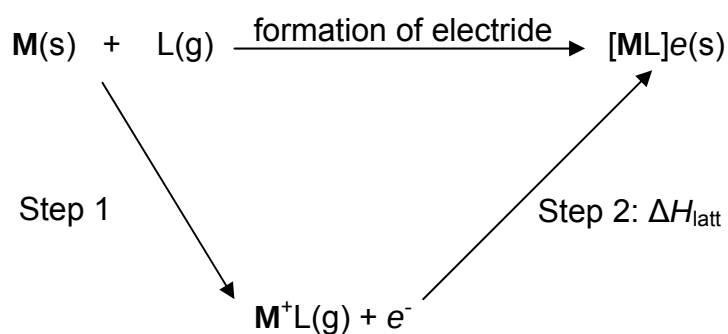
[5]

[Total: 10]

- 5 An electride is a special type of ionic compound that can be formed from Group I metals (such as Na and Cs) and has 'trapped' electrons taking on the role of anions. It has the general formula $[\text{ML}_n]\text{e}^-$, where M, L and e refer to the metal, ligand and electron respectively. For example,



A generalised energy cycle for the formation of electrides is shown below.

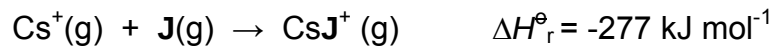


- (a) (i) Define, with an aid of an equation, what is meant by the first ionisation energy of a metal **M**.
- (ii) Step 1 is proposed to be the determining factor in whether the formation of the electride is possible.

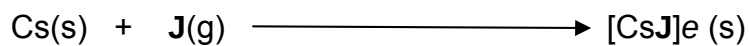
Based on this, which of the group I metals of sodium or caesium is predicted to form electrides most readily? Explain your answer.

- (ii) **J** is a hexadentate ligand, $(\text{CH}_2\text{CH}_2\text{OCH}_2\text{CH}_2\text{OCH}_2\text{CH}_2)_3\text{N}$ that can be used in the formation of an electride.

When the ligand **J** binds to a Cs^+ ion, the enthalpy change for the reaction is as follows:



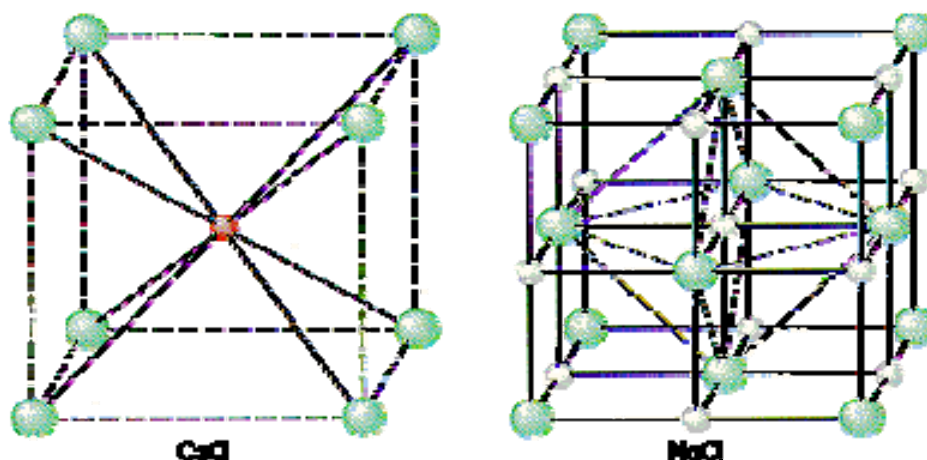
Using the additional data given in the table below, calculate a value for the enthalpy change of formation of the electride, $[\text{CsJ}]\text{e}(\text{s})$, defined as follows:



enthalpy change of atomisation of Cs	+77 kJ mol ⁻¹
first ionisation energy of Cs	+378 kJ mol ⁻¹
lattice energy of $[\text{CsJ}]\text{e}$	-292 kJ mol ⁻¹

[7]

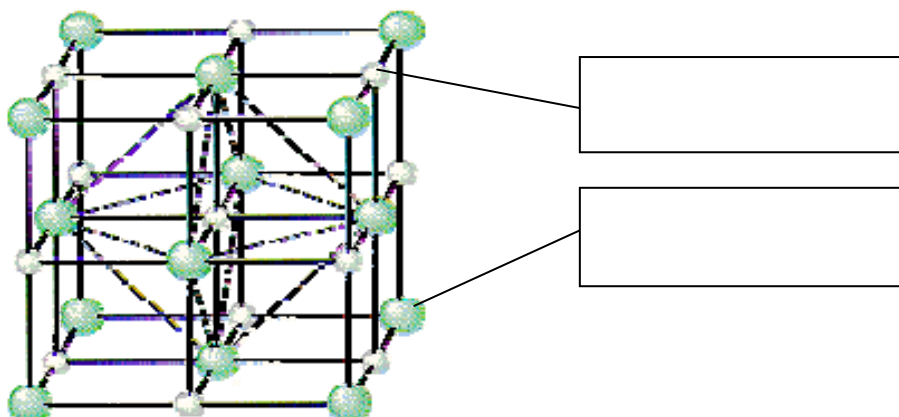
The structures of some electrides are studied and they are said to be remarkably similar to that of sodium and caesium chlorides. The diagram below shows the crystal lattice structures for both sodium chloride and caesium chloride.



crystal lattice of CsCl and NaCl

In the crystal lattice of sodium and potassium chloride, the co-ordination number of each ion is 6. However, in the crystal lattice of caesium chloride, CsCl, the co-ordination number has a different value.

- (b) (i) From the diagrams, deduce the co-ordination number in CsCl lattice. Suggest an explanation for the co-ordination number in the CsCl lattice being different from those in NaCl and KCl.
- (ii) It is proposed that $[\text{Na}(\text{NH}_3)_6]\text{e}$ has the same crystal lattice structure as NaCl. In the diagram below, label the particles present in the structure.



- (iii) Unlike Na which forms $[\text{Na}(\text{NH}_3)_6]\text{e}$, lithium forms the electride $[\text{Li}(\text{NH}_3)_4]\text{e}$ instead.

Suggest a reason why this is so.

- (iv) Suggest whether an electride can conduct electricity in the solid state. Explain your reasoning fully.

[5]

[Total: 12]

END OF PAPER

